

Proxy Load Test Report: Auto-Scaling and Performance Analysis

Proxy Load Test Report — May 7, 2026

Summary

Auto-scaling validated. The proxy handles 1,367 RPS with zero errors, auto-scales under load, and maintains P95 < 340ms throughout.

Test Configuration

Setting	Value
Proxy architecture	Per-backend upstream pools, proxy_ssl_session_reuse on
ECS tasks	2 × 0.5 vCPU / 1GB (ARM64 Graviton)
Auto-scaling	Min 2, Max 20, CPU target 70%
mTLS	Enabled (client cert presented on every request)
Payload	9,681 varied availability requests (circular feeder)
Connection model	Concurrent (persistent connections, realistic)

Scaling Test Results (220 concurrent users)

Metric	Value
Total requests	2,048,135
Errors	0
Mean RPS	1,367
Peak RPS	~1,500
Mean response	129ms

P95	336ms
P99	470ms

CPU & Scaling Behaviour

Phase	CPU (avg)	Tasks	Notes
Ramp (0-5 min)	Rising to 70%	2	Load increasing
Threshold hit	80%	2	Scaling triggered
Scale-out 1	Dropping	3	First new task added
Scale-out 2	45%	4	Second new task added
Steady state	45%	4	Load distributed

Key Findings

1. **Proxy handles 1,367 RPS with zero errors** — well above expected production load
2. **Auto-scaling works correctly** — triggered at 70% CPU, added 2 tasks, CPU dropped to 45%
3. **P95 maintained at 336ms** — no degradation during scaling
4. **mTLS overhead negligible** — with persistent connections, TLS handshake only happens once per connection

Previous Test Comparisons

Test	Users	RPS	CPU	P95	Errors
Baseline (7 users)	7	119	9%	100ms	0
Varied payloads (60 users)	60	314	30%	336ms	0
Scaling attempt	140	855	55%	341ms	0

(140 users)					
Scaling triggered (220 users)	220	1,367	80%→45%	336ms	0

Conclusion

The proxy is production-ready. At realistic traffic levels (persistent connections, mTLS), it handles well above expected production TPS with headroom. Auto-scaling provides additional capacity when needed.